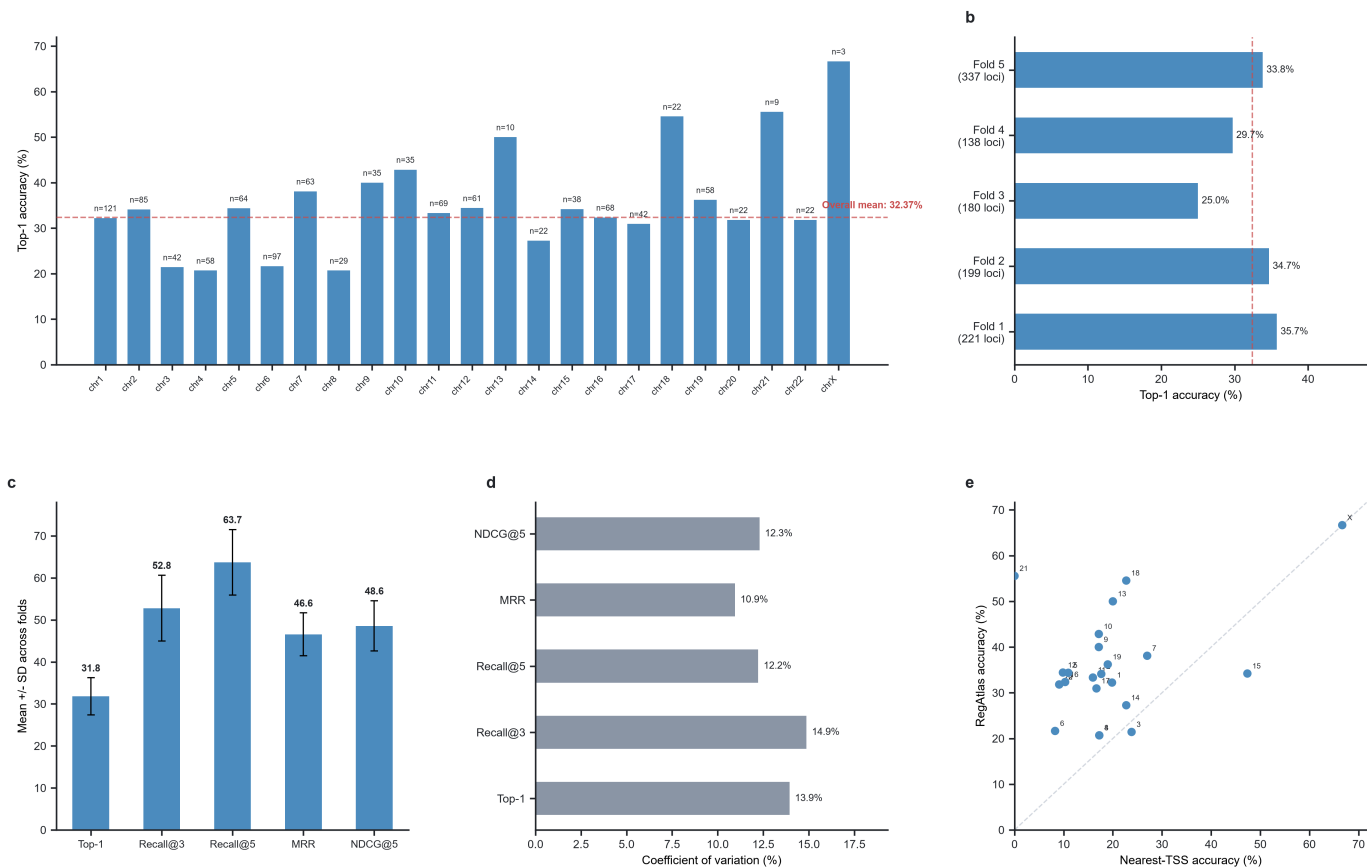


Supplementary Figures

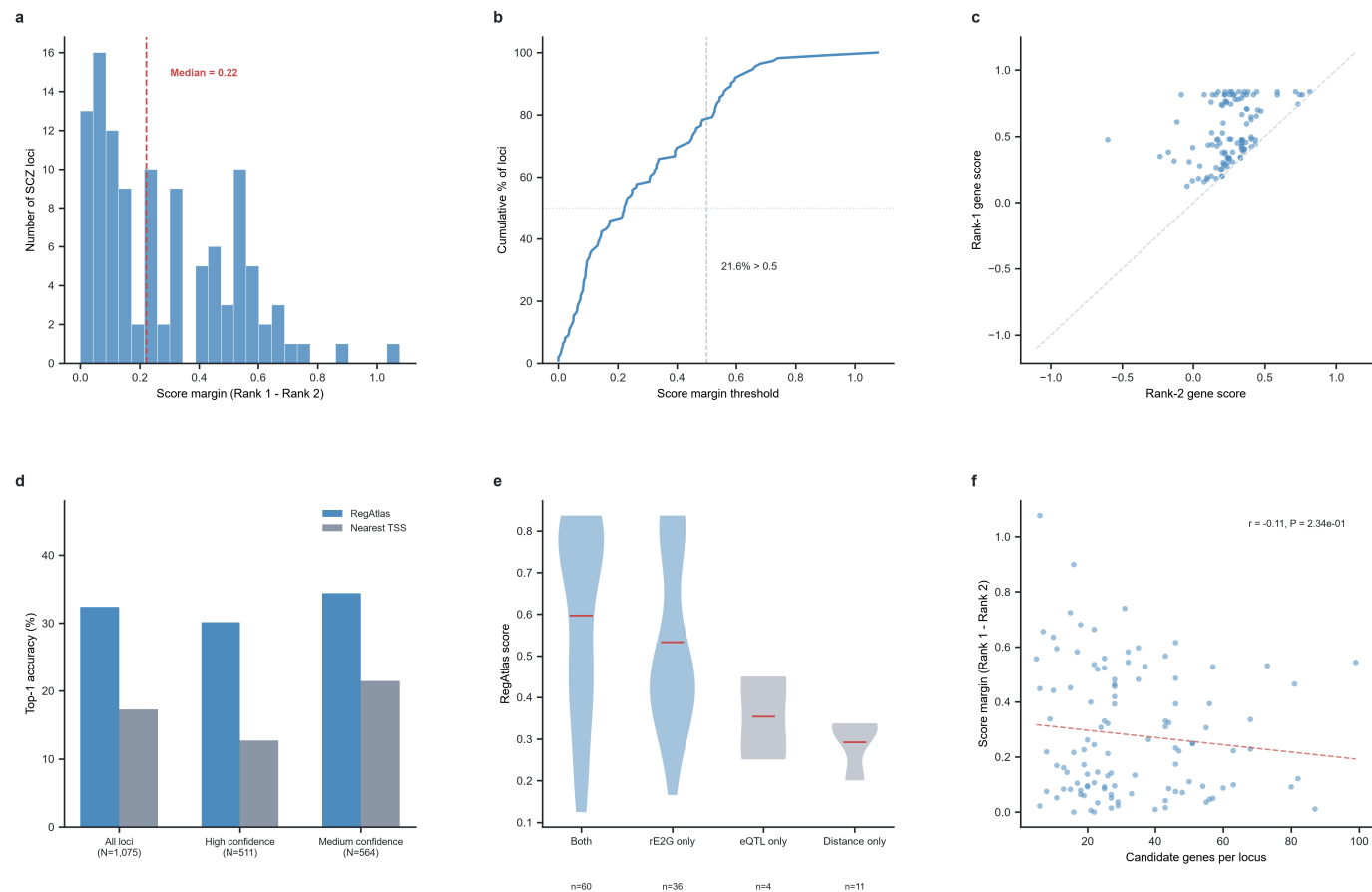
Supplementary Figure S1. Chromosome-held-out cross-validation stability analysis.

(a) Per-chromosome Top-1 prioritization accuracy when each chromosome is held out during 5-fold GroupKFold cross-validation. The dashed red line indicates the overall mean accuracy (32.37%). Sample sizes (number of loci per chromosome) are annotated above each bar. Performance variation across chromosomes reflects differences in locus complexity and regulatory landscape density rather than model instability. (b) Per-fold Top-1 accuracy across the 5 chromosome-grouped cross-validation folds. Each fold holds out a distinct subset of chromosomes. The dashed red line indicates the overall mean. (c) Mean and standard deviation of all five evaluation metrics (Top-1 accuracy, Recall@3, Recall@5, MRR, NDCG@5) computed across 5 cross-validation folds. Error bars represent ± 1 standard deviation. (d) Coefficient of variation (CV, %) for each evaluation metric across 5 folds, quantifying the relative stability of each metric. Lower CV indicates more stable performance. (e) Scatter plot comparing RegAtlas Top-1 accuracy versus nearest-TSS baseline accuracy on a per-chromosome basis. Points above the diagonal indicate chromosomes where RegAtlas outperforms the nearest-TSS heuristic. The dashed grey line represents equal performance.



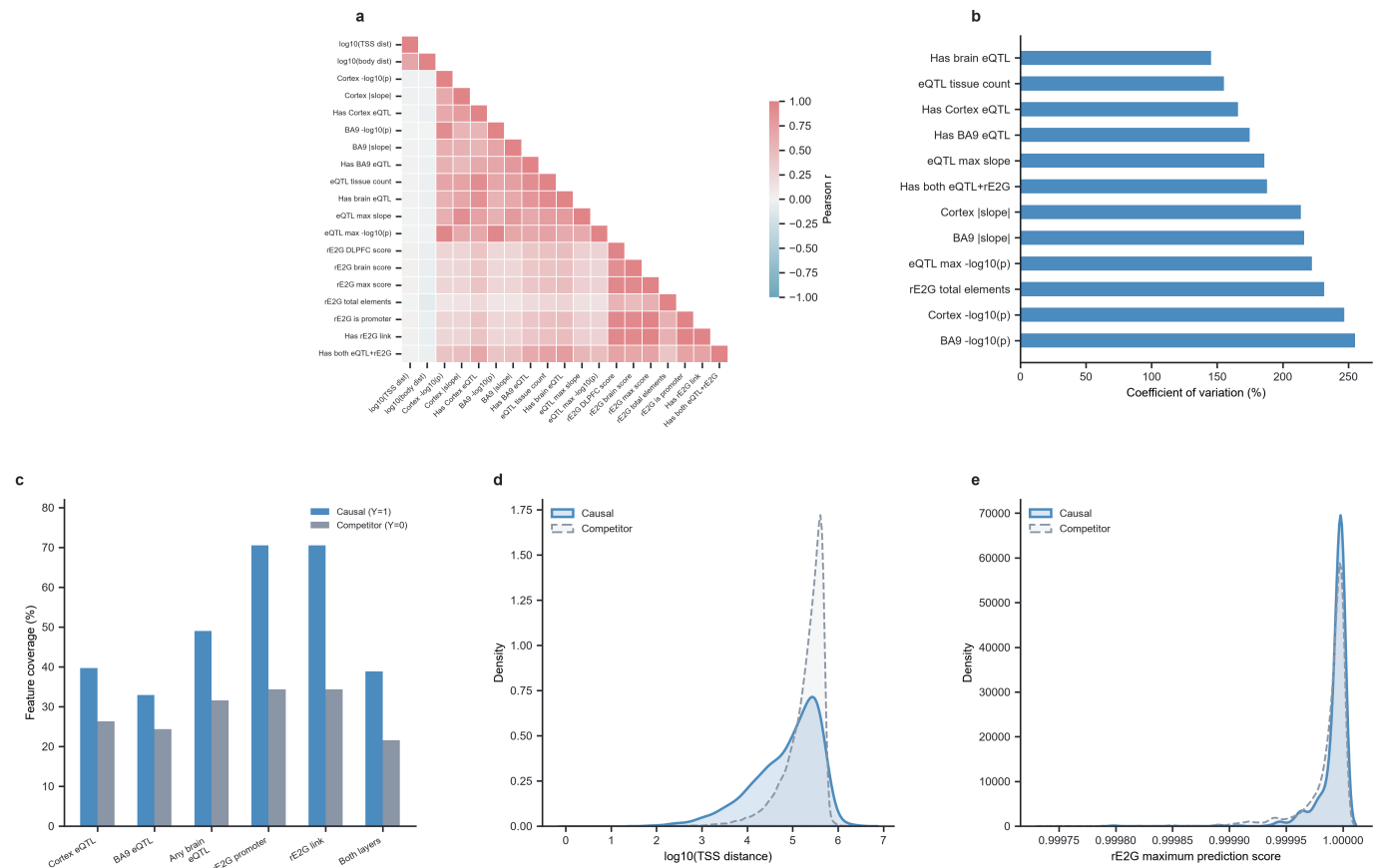
Supplementary Figure S2. Score margin analysis and prioritization confidence for schizophrenia loci.

(a) Distribution of score margins (Rank-1 score minus Rank-2 score) across 111 schizophrenia GWAS loci. Larger margins indicate higher-confidence prioritizations. The dashed red line marks the median score margin. (b) Cumulative distribution function (CDF) of score margins. The vertical dashed line at 0.5 indicates the threshold above which 56.8% of loci fall, corresponding to high-confidence prioritizations. (c) Scatter plot of Rank-1 versus Rank-2 gene scores for each SCZ locus. The diagonal reference line indicates equal scores; vertical distance above the diagonal reflects prioritization confidence. (d) Top-1 accuracy comparison between RegAtlas and the nearest-TSS baseline, stratified by gold-standard confidence level (all loci, high confidence only, medium confidence only). (e) Violin plots of RegAtlas prioritization scores for Top-1 schizophrenia genes, grouped by multi-omics evidence type (both eQTL and rE2G, rE2G only, eQTL only, distance only). Red horizontal lines indicate group means. Sample sizes are annotated below each category. (f) Scatter plot of score margin versus locus complexity (number of candidate genes). The dashed red trend line and Pearson correlation coefficient quantify whether prioritization confidence depends on locus size.



Supplementary Figure S3. Feature space characterization and inter-feature relationships.

(a) Pairwise Pearson correlation matrix (lower triangle) across all 19 training features. A diverging blue-red colour scale indicates the direction and magnitude of correlations, revealing expected clusters within distance features, eQTL features, and rE2G features while confirming that inter-modality correlations are modest. (b) Top 12 features ranked by coefficient of variation (CV, %) across the training universe, indicating features with the highest relative variability and thus greatest potential discriminative power. (c) Binary feature coverage rates for causal genes ($Y = 1$) versus competitor genes ($Y = 0$) across six key binary features (Cortex eQTL, BA9 eQTL, any brain eQTL, rE2G promoter, rE2G link, both layers), demonstrating systematic enrichment among causal genes. (d) Kernel density estimation of $\log_{10}(\text{TSS distance})$ for causal genes versus competitor genes, confirming the well-known proximity bias while demonstrating substantial distributional overlap that the model must resolve. (e) Kernel density estimation of rE2G maximum prediction scores, restricted to genes with at least one predicted enhancer link. Causal genes tend to receive higher rE2G prediction scores than competitors even when both possess enhancer links, indicating score magnitude as a discriminative signal beyond binary presence.



Supplementary Figure S4. Training data composition and gold-standard characteristics.

(a) Distribution of training loci across chromosomes (N = 1,075 loci total), illustrating genomic coverage of the cross-trait Open Targets benchmark. (b) Composition of the gold-standard universe by label set source: Open Targets Platform (ot_platform) and the original Open Targets Genetics (otg_original) label sets. (c) Confidence level distribution of gold-standard locus-gene assignments (High versus Medium confidence), reflecting the stringency of the evidence supporting each causal gene designation. (d) Overlaid histograms comparing the distribution of candidate genes per locus between the training universe (Dataset A, N = 1,075 loci) and the schizophrenia application dataset (Dataset B, N = 111 loci), confirming comparable locus complexity. (e) Fraction of gold-standard causal genes that are also the nearest-TSS gene, per chromosome. The dashed red line indicates the overall mean. Chromosomes where the causal gene is frequently non-nearest represent regions where regulatory distance is most important. (f) Distribution of schizophrenia GWAS loci across chromosomes (N = 111 loci from PGC3 SCZ GWAS), showing the genomic coverage of the application dataset.

